

1) Write a Python program that creates a list of 5 integers and performs the following tasks:

- a) Prints the first and last element
- b) Adds a new number to the list
- c) Removes the second element

2) Create a tuple colors = ("red", "green", "blue"). Print the second element. Try to change "green" to "yellow", What happens?

3) Given person = ("Alice", "25", "Berlin"), unpack the tuple into separate variables and print them in a sentence

"Alice is 25 years old and lives in Berlin"

4) Write a program that creates 2 sets: A = {1,2,3,4} and B = {3,4,5,6}. Find and print: Union of A and B, Intersection of A and B and the Set Difference of A and B

5) Create a set with duplicate values. Print the set, What do you notice?

6) Create a dictionary with student names as keys and their scores as values. Print a person's score using their key. Add a new student "David" : 92 and update a random score of anyone to 88

7) Given the Dictionary capitals = {"Germany": "Berlin", "France": "Paris", "Italy": "Rome"}, write a program that prints each country and its capital in the format : "The capital of Germany is Berlin"

9) Write a Python program that stores a list of dictionaries representing students:

```
students = [  
    {"name": "Alice", "age": 20},  
    {"name": "Bob", "age": 22}  
]
```

Print the names of all the students in the list

10) Write a Python program that:

- Stores a sentence in a string.
- Splits it into words (list).
- Converts the list into a set to remove duplicates.
- Stores each unique word as a key in a dictionary, with the value being the word length.

11) Given a list of numbers numbers = [10,20,30,40,50], write a script that prints each element using a loop, Calculate the sum of all elements without using sum()